Junyi (Conny) Zhou

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Education

- **Harvard University, T.H. Chan School of Public Health** — Boston, MA
  S.M. in Health Data Science — *August 2025 – Expected December 2026*
  - Coursework: Deep Learning, Healthcare Machine Learning
- **Emory University, College of Arts and Sciences** — Atlanta, GA
  B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925 — *August 2021 – May 2025*
  - Coursework: Data Structures, Database Systems, Machine Learning, Natural Language Processing

Technical Skills

- **Languages**: Python, C++, Java, JavaScript, SQL, R
- **Software**: Flask, React, REST APIs, Auth0, Git, MongoDB, PostgreSQL, MySQL
- **AI/ML**: PyTorch, OpenAI APIs, RAG, scikit-learn, Hugging Face Transformers
- **Cloud & Tools**: AWS Elastic Beanstalk, CloudFront, EC2, S3, Jupyter, Cursor

Software Engineering Experience

- **Airway Management Simulation Platform** — Atlanta, GA
  Full-Stack Engineer, Scrum Master — Emory Pediatric Hospital — *January 2024 – Present*
  - Architected a **React** client and **Python/Flask REST API** backed by **MongoDB**, with **Auth0** authentication for a clinical training platform.
  - Deployed the Flask backend through **AWS Elastic Beanstalk** and delivered the frontend through **CloudFront**, configuring SSL and DNS for browser access.
  - Integrated an **OpenAI RAG pipeline** with context management and hallucination controls to generate multi-turn simulation scenarios.
  - Led an Agile team of 6, conducting code reviews and incorporating clinician feedback to iteratively improve the application and NLP pipeline.

Technical Projects

- **Transformer Architecture Reimplementation** — Atlanta, GA
  Machine Learning Researcher — Emory Dept. of Computer Science — *October 2024 – January 2025*
  - Implemented core Transformer algorithms from first principles in **PyTorch**, including multi-head self-attention, positional encoding, layer normalization, and the training pipeline.
  - Tuned and benchmarked the implementation against `nn.Transformer`, validating correctness and achieving a **9.38 BLEU** score on English-to-German translation.
- **Parallel AlphaFold Inference Pipeline** — Atlanta, GA
  Data Analyst — Bou-Nader Lab, Emory Dept. of Biochemistry — *February 2025 – August 2025*
  - Automated high-throughput inference on HPC clusters, using parallel scheduling to distribute structural predictions across GPU nodes.
  - Built **Python** scripts and CLI tools to filter and rank outputs using pLDDT and PAE scores, with documentation for non-ML researchers.
- **CellOT Cross-Species Modeling** — Boston, MA
  Graduate Researcher — Mooney Lab & Alvarez-Melis Lab, Wyss Institute — *February 2026 – Present*
  - Implemented a custom **PyTorch CellOT** model and **scanpy** data pipelines for latent-space modeling, automated labeling, and single-cell dataset integration.